Multimodal fusion of clinical time-series data, chest X-ray images and medical notes for healthcare applications

Alireza Ghasemian Moghadam



Department of Computer Engineering
Ferdowsi University of Mashhad
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Supervisor: Prof. Hadi Sadoghi Yazdi

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Abstract

In the realm of healthcare, the fusion of multimodal data holds immense potential for enhancing clinical decision-making processes. While conventional machine learning approaches often focus on a single modality, this study introduces a novel neural network architecture capable of integrating multimodal patient data—specifically, free-text clinical notes, chest radiographs and clinical data from the publicly available MIMIC-IV database. Through a retrospective analysis involving 6,125 intensive care patients, our model demonstrates superior performance in predicting life expectancy and disease classification compared to models relying solely on any single one of the modalities or combinations of them. Furthermore, we establish the robustness of our approach, particularly in scenarios where complete clinical data may not be available. This integrated approach not only advances our understanding of multimodal learning in healthcare but also provides a standardized framework for assessing and advancing machine learning applications in clinical settings.

Key Words

Life Expectancy, Medical Information Mart for Intensive Care, MIMIC, Survival Prediction, Phenotype classification, Deep Learning, Intensive Care, Multimodal Learning

1 Introduction

Intensive care units (ICUs) play a vital role in modern medicine, accommodating seriously ill patients and consuming approximately 1% of the US GDP annually (Halpern and Pastores, 2010). With the rise of digital records, vast amounts of clinical data are amassed during ICU stays. Concurrently, over 30 million US hospital visits annually contribute to a rich source of electronic health records (EHRs), utilized by 83% of patients (Henry et al., 2016). This data reservoir offers ample opportunities for machine learning and data mining research, addressing healthcare challenges like early triage, risk prediction, and disease characterization (Stahlschmidt et al., 2022; Mohsen et al., 2022; Huang et al., 2020a; Behrad and Abadeh, 2022). While these issues are longstanding, the advent of machine learning (Mnih et al., 2013) coupled with the abundance of clinical data has intensified interest in their resolution. In most applications (Stahlschmidt et al., 2022), only structured clinical data is used, while clinicians make use of data such as imaging and clinical notes in their decision making process.

Deep learning models excel in medical tasks, often surpassing specialist clinicians (Spasov et al., 2018). Integrating Electronic Health Records (EHR) with imaging data enhances diagnosis prediction (Hayat et al., 2023). However, current methods underutilize text data. Combining clinical notes with other modalities can boost automated medical analysis (van Sonsbeek and Worring, 2020). Multimodal (every source of data is a modality) learning, common in audio-visual fields (Akbari et al., 2021), is less explored in healthcare. It aims to enhance task performance by leveraging diverse data sources, a strategy crucial for advancing medical imaging and diagnosis. Consequently, there's a demand for machine learning models that seamlessly integrate clinical, textual, and imaging data to optimize healthcare outcomes though such datasets are hard to find and access (Soenksen et al., 2022).

One of the publicly and comprehensive datasets for multimodal learning in healthcare is MIMIC (Medical Information Mart for Intensive Care) (Johnson et al., 2019a,b, 2023b,a). It contains de-identified health data from patients admitted to critical care units at Beth Israel Deaconess Medical Center in Boston, Massachusetts, USA. The dataset includes information such as demographics, vital signs, laboratory tests, medications, procedures, chest X-Rays, electrocardiogram, echocardiogram and clinical notes.

Little work has been done on MIMIC where data from all these modalities or combinations are used and comparisons between them made. In this work, we integrate data from the three main data modalities and compare performance with baselines and results from other publications. We also utilize a multimodal dropout scheme where data from each modality is dropped at training time to help with generalization to real world scenarios with missing data. The rest of the paper is structured as follows: related work and background are explored in Section 2. The proposed approach is explained in Section 3. Section 4 presents the results and comparisons with other methods. Section 5 features discussions about the paper's overarching findings and the paper's conclusion.

2 Background

2.1 Literature Review

2.1.1 Search Criteria

For this review we used the guidelines from PRISMA (Tricco et al., 2018). The focus of this review is on the application of machine learning and deep learning to multimodal healthcare tasks, in which multimodal is defined as having more than two data modalities including structured clinical data, unstructured free-text data and also medical images. To obtain the set of related articles, we first searched in Institute of Electrical and Electronics Engineers (IEEE) Digital Library, PubMed, google scholar, and Association for omputing Machinery (ACM) Digital Library with query terms in the following format: ("multimodal" AND ("machine learning" OR "deep learning") AND ("EHR" OR "imaging" OR "text")). We used various forms of each keyword, and included different ML and DL algorithms. The search resulted in 220 studies. Our inclusion criteria are that the research study must be using the aformationed modalities and cannot be using only one of the modalities in the proposed model, and that the model must use ML or DL techniques. Based on these criteria, we screened 103 articles according to the title and abstract at the first step, removed duplicates, and after full-text review, 42 articles were left. From refrences and using google, we then snowballed on other relevant review articles and included studies. Our final pool has 55 studies. During full text review, we analyzed the studies from three perspectives: data, task, and model. For data, we focused on the dataset used by the studies, including size, availability and data types used (categorical, numerical, free-text, etc.). For task, we looked at where and how the proposed model is applied, and recorded elements such as task type, disease domain, and outcome. Finally, for the model, we focused on how each modality is represented, categorized the fusion strategy of the modalities, and recorded implementation details of the model.

2.1.2 Related Literature

In a landmark paper, Harutyunyan et al. (2019) proposed the four public benchmarks mortality prediction, decompensation prediction, length-of-stay prediction, and phenotyping for the MIMIC-III dataset to evaluate and compare research papers and machine learning methods to help facilitate progress in healthcare research. They developed linear regression models and two LSTM-based neural baselines (standard LSTM and channel-wise LSTM) for the experiments. Additionally, they tested both types of LSTMs with deep supervision and multitask training. In deep supervision, the model has to have to make multiple predictions and we supervise the model at each time step. In multitask training we train the model on all four tasks at the same time. The LSTM-based models outperformed linear models across all metrics on every task. The results also showed that channel-wise LSTMs perform significantly better than standard LSTMs for all four tasks, while multitasking helps for all tasks except phenotyping, and channel-wise LSTMs and multitask training act as regularizers. The results of deep supervision were mixed with it not helping in some cases.

Instead of developing a separate model for each modality Kawahara et al. (2019) experiemented with a multimodal convolutional neural network with a multi-task loss function and trained the pipeline in a single optimiztion which also helped with the case of missing data for modalities.

Frid-Adar et al. (2018) and Waheed et al. (2020) both used GANs to make synthetic medical image data and both found that the model trained on both real and synthetic data performs better than model trained only on real data.

Like Harutyunyan et al. (2019), Purushotham et al. (2018) also presented benchmarks for Mortality prediction, ICD-9 code group prediction, and Length of stay prediction. The mortality prediction task was divided into in-hospital mortality prediction, short-term (2-day and 3-day), and long-term (30-day and 1-year). For prediction, they used Scoring methods such as Simplified Acute Physiology Score II and Sepsis-related Organ Failure Assessment, Super Learner models, and DL with three feature sets consisting of raw or processed data from patients' ICU stays being fed as inputs. They observed that DL models outperform all other methods given a large number of raw clinical time series data is used as input features.

In all of these works, the patients were analyzed as a whole with a single, global model. In contrast, Suresh et al. (2018) used divided cohorts learned in an unsupervised manner in a multi-task framework which showed better performance than the global setting.

Waudby-Smith et al. (2018) showed that nursing staff's sentiments from clinical notes have a statistically significant correlation with 30-day mortality and can be used as a feature for a classifier. Results from Gao et al. (2021) on MIMIC-III and Kang et al. (2022) on their private dataset corroborate this.

Lee et al. (2019) used a GRU for feature learning from every data modality, then simply concatenated feature vectors and fed it to linear regression for classification and regression. Their reason for LR was that it provides good interpretability. They experimented with both real (ADNI) and simulated data and found better performance for multimodal setting than unimodal setting.

Cheerla and Gevaert (2019) used a cosine similiarity loss to train a multimodal network for predicting pancancer prognosis, where for the views from the same training examples, the cosine similarity between these feature representations is maximized, whereas for views from different examples, the cosine similarity is minimized. To ensure robustness in the face of missing data modalities, they proposed multimodal dropout where at training time feature vectors corresponding to each modality is dropped and the weights of the other modalities is caled up.

Alsentzer et al. (2019) trained BERT on MIMIC-III notes and released their models as clinicalBERT.

Taleb et al. (2021) for example mixed four image modalities using permutations and then had the network try to learn uncorrupted combination.

van Sonsbeek and Worring (2020) used MIMIC-CXR using both imaging and text modalities in an intermediate fusion approach to classify diagnosis and generate free-text medical reports. In both tasks multimodal setting performed better than unimodal approaches and only EHR approach performed better than only images approach.

Huang et al. (2020a) literature review shows that additional patient information and clinical context can result in better model performance and fusion methods better replicate the human expert interpretation workflow, it is recommended to always experiment with fusion strategies when multimodal data is available.

Similarly, Huang et al. (2020b) also studied 7 different fusion architectures with multimodal models outperforming single-modal ones but different fusion models performed better for different criteria.

Li et al. (2020) examined four pre-trained vision and langauge transformer-based models and found that learning joint image-text embedding are superiour to only learning text embedding and that models also outperform CNN-RNN models.

To learn better intermediate features for joint fusion, Joze et al. (2019) proposed a new module based on the squeeze and excitation where by adding the module to different levels of the unimodal branches of models, we can learn fused knowledge which in turn improves the performance. Their work

Meng et al. (2021) found that mortality prediction deep learning models focus unequally on the demographic features of patients for prescribing medication, raising questions about the fairness of these models. They also discovered that models look beyond proven clinically important features and recognize new important ones for prediction.

Ho et al. (2021) also trained multiple shallow ML Models and deep learning models, notably they used transformer, temporal convolution, and auto-encoder networks to predict colorectal cancer recurrence using multimodal data. They found hybrid models to perform better overall.

Jiao et al. (2021) trained two models to predict severity and progression of COVID-19 using both chest x-rays and clinical data. For severity the x-rays went through a conv net and clinical features through a FFN while for progression features from both modalities went through random survival forests. Their work can be classified as late fusion and the modalities were combined by weighted sum.

Hayat et al. (2023) proposed MedFuse, a multimodal fusion approach to aggregate representations from both MIMIC-IV and MIMIC-CXR beating the state of the arts in in-hospital mortality prediction and phenotype classification. First, they trained a ResNet-34 on chest X-ray images and an LSTM network on the same variables as Harutyunyan alone for each prediction task for pre-training. They then concatenated latent representations from two networks, passing them through another LSTM network and training again. Grant et al. (2021) used the same methodology for the classification of cardiomegaly.

To effectivly fuse medical images and EHR, Jiang et al. (2021) used two attention based modules to combine and complement modalities with both initiatives increasing accuracy.

Liu et al. (2021) in their systematic review identefies MIMIC as the most popular public dataset for multimodal machine learning analysis.

Akbari et al. (2021) trained one transformer (shared between modalities or for every modality but both cases with the same architecture) in a self-supervised manner and then fine tuned on different tasks and saw that it performed better then simply training in a supervised manner.

Walston et al. (2022) tried to predict whether a patient require mechanical ventilation, or require ICU admission or was expected to die using clinical data and Chest radigraphs. Their model can be classified as early or intermediate fusion. The multimodal model performed better in all 3 tasks then unimodal models.

Xie et al. (2022) proposed MIMIC-IV benchmarks and an open-source pipeline for emergency department (ED) related tasks. The hospitalization outcome, The critical outcome, and ED reattendance were introduced as three main tasks, and Various traditional and deep learning models were tested. They also found that complex deep learning modes did not perform better than simpler, more interpretable methods.

Gupta et al. (2022) developed an end-to-end customizable pipeline for MIMIC-IV to help with cohort extraction, preprocessing, modeling and evaluation, and fairness analysis. Both ICU and non-ICU data were taken into account and all prediction tasks like previous studies were incorporated.

Ma et al. (2022) investigated the robustness of mutlimodal transformers in face of missing modality in test data. They also found that to fusion strategies are dataset dependent.

Similar to Gupta et al. (2022), Soenksen et al. (2022) proposed Holistic AI in Medicine (HAIM) pipeline to process all four modalities of MIMIC-IV and make the data ready for modelling. To extract features from vision and text modalities, SOTA models like Clinical BERT and DENSENET-CXR were used, Their work can be classified as early fusion because features are extracted and then the features are used for modelling.

Using the MIMIC-IV note (Johnson et al., 2023a), Nguyen et al. (2023) expanded upon

previous work for MIMIC-III (Mullenbach et al., 2018, Shiet al., 2017) and introduced benchmarks for ICD-10 and ICD-9 code prediction from clinical notes, testing different models on the new dataset and reporting the results.

Wang et al. (2023) explored the use of an explainable machine learning model for predicting the duration of mechanically ventilated patients in MIMIC-IV and eICU-CRD and found XGBoost to be the most stable and accurate model.

Following Hayat et al. (2023), Khader et al. (2023) used clinical data in combination with imaging data for survival prediction. They used the transformer architecture and reached the same conclusions.

Hsieh et al. (2023) used clinical data in conjunction with chest X-rays to detect regions in the X-ray images with abnormalities more accurately. The dataset was made from combining MIMIC-ED and MIMIC-CXR-JPG and their model was based on Mask R-CNN with some changes to accommodate clinical data. The addition of clinicla data improved the results Average Precision by 12%. THey also found that element-wise sum combining of modalities perform better than other methods in this application.

Zhang et al. (2023) introduced meta-transformer, a unified transformer-based model taking in data from 12 modalities, tokenizing them and projecting them into a shared space, encoding and extracting features with a shared-weight transformer and then using task specific heads for downstream tasks.

To read more on these topic, we refer the reader to these comprehensive reviews: Stahlschmidt et al. (2022); Mohsen et al. (2022); Huang et al. (2020a); Behrad and Abadeh (2022).

2.2 Open Challenges

Reading through the literature we identified the following gaps:

1. No work has been done like Ma et al. (2022) where different medical multimodal datasets are taken and different fusion strategies are applied and compared. There are different ways to do fusion and also different ways to combine features from modalities whether through early, joint or late fusion. For example to fuse predictions from late fusion models only simple methods have been used whereas there are a lot of methods available (Rokach, 2010).

2. Works like Hayat et al. (2023); Khader et al. (2023); Soenksen et al. (2022) though incorporating lots of modalities, still not use all available modalities of the datasets.

3. Similar to Akbari et al. (2021); Zhang et al. (2023) No work has been done to use transformers for all modalities of the datasets.

4. Using self-supervised and unsupervised learning techniques for pre training models have not been explored (Akbari et al., 2021; Zhang et al., 2023; Alayrac et al., 2020)

In this work, we will take Hayat et al. (2023) work and add text modality to the models, apply the models to all tasks and compare preformance.

3 Methodology

3.1 Dataset

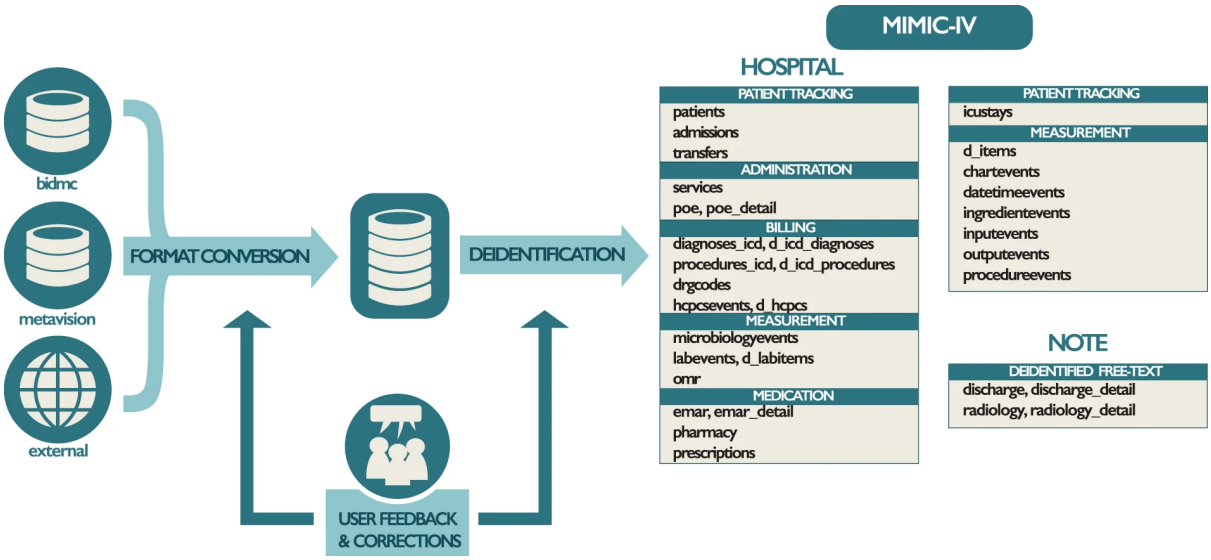


Figure 1: An overview of the development process for MIMIC. Data are acquired from the BIDMC data warehouse, the ICU information system (MetaVision), and external sources ("acquisition"). Structured Query Language (SQL) scripts merge the various data sources into a single schema ("transformation"). Finally, deidentification algorithms are applied to selectively remove protected health information from the reformatted schema. Tables present in MIMIC-IV are provided on the far right of the figure under their respective module. Issues raised on the MIMIC Code Repository are assessed and used to improve the build process as appropriate. (From https://www.nature.com/articles/s41597-022-01899-x)

This study used the Medical Information Cart for Intensive Care (MIMIC)-IV (Johnson et al., 2023b), a publicly available database containing de-identified health records of 383,220 patients admitted to the intensive care unit (ICU) or emergency department (ED) of Beth Israel Deaconess Medical Center (BIDMC) in Boston, Massachusetts, USA, between 2008 and 2019. MIMIC-IV's latest version (v2.2) builds upon MIMIC-III to grant public access to electronic health record (EHR) data for over 40,000 hospitalized patients, culled from BIDMC's MetaVision clinical information system. We opted for MIMIC-IV due to its extensive scope, in-depth documentation, standardized formatting, established use in AI/ML applications. We incorporated two other datasets into MIMIC-IV v2.2. First the MIMIC Chest X-ray (CXR) database v2.0.0 (Johnson et al., 2019a,b), which harbors 377,110 radiological images with free-text reports encompassing 227,835 medical imaging events that can be linked to corresponding patients within MIMIC-IV. Second MIMIC-IV-Note (Johnson et al., 2023a) containing 331,794 deidentified discharge summaries from 145,915 patients admitted to the hospital. MIMIC-IV-Note also contains 2,321,355 deidentified radiology reports for 237,427 patients overlapping with MIMIC-CXR. There are other parts in MIMIC data as well such as MIMIC-ECHO (Gow et al., a) and MIMIC-ECG (Gow et al., b) which we didn't use.

To read more about MIMIC Dataset, read the documentation at https://mimic.mit.edu/docs/

Table 1: Characteristics of the combined dataset. (From Khader et al. (2023) - No missing data)

Parameter	All patients	Training set	Validation set	Test set
No. of paired chest radiographs and clinical parameters	6798	4885	540	1373
No. of patients	6125	4396	472	1257
No. of male patients	3382 (55%)	2414 (55%)	264 (56%)	704 (56%)
Age (years) (range)	18—91	18—91	19—91	18—91
Mean age ± SD [Median] (years)	64 ± 16 [65]	64 ± 16 [65]	64 ± 16 [65]	64 ± 16 [65]
No. of deaths during hospital stay	1002 (16%)	717 (16%)	76 (16%)	209 (17%)

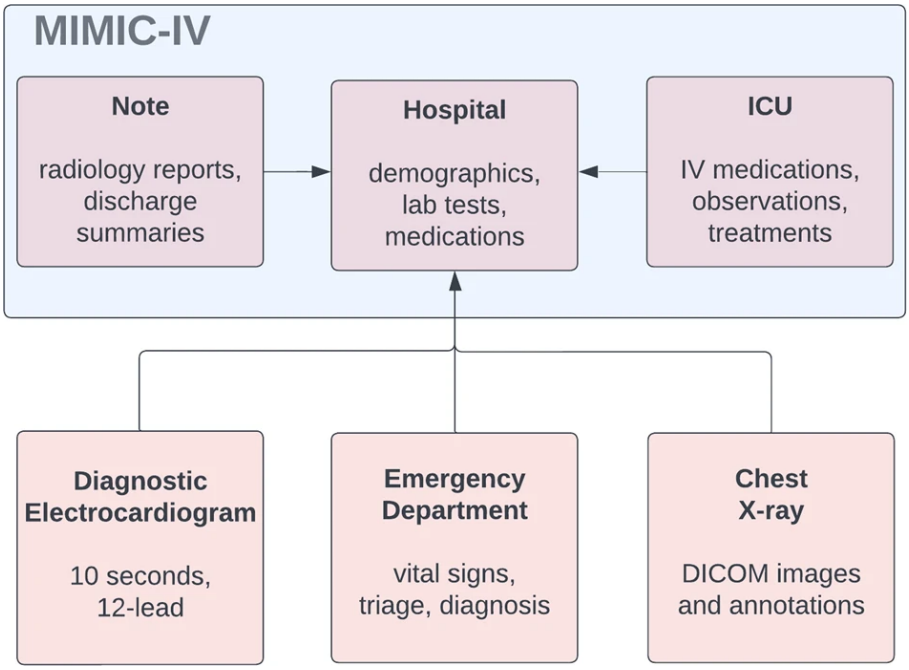


Figure 2: MIMIC-IV follows a modular structure. Modules can be linked by identifiers including subject_id, hadm_id, and deidentified date and time. Example content of each module is shown. (From https://www.nature.com/articles/s41597-022-01899-x)

3.2 Proposed method

We extend the work of Hayat et al. (2023) to three modalities adding free-text clinical notes. The approach consists of two stages. In the first stage, we train separate models, each tailored to a specific modality, to extract features from that modality. In the second stage, we combine these modality-specific features using a multimodal fusion module, allowing them to interact and inform each other. The artitecture is pretty straightforward. For every modality, there is a modality specific encoder, specialized for data from that modality.

For chest X-Ray images we use the feature extractor of a ResNet-34 (He et al., 2015) pretrained on CheXpert (Irvin et al., 2019) (another chest X-Ray dataset) which gives us an output with the size of $n = 512$. For clinical time-series data we use two stacked layers of an LSTM (Hochreiter and Schmidhuber, 1997) network with a dropout (Srivastava et al., 2014) layer giving us an output size $m = 256$. For clinical text notes first we pass the data through BioWordVec (Zhang et al., 2019) (a model pretrained on publicly available biomedical corpora,

such as PubMed abstracts and clinical notes from the MIMIC-III dataset) to get embeddings. We then pass the embedding through another two layer LSTM network with the final size being $t = 256$. Each of these modality specific models are trained seperately for every task by applying a linear classifier on top of them and training them with Binary Cross-Entropy loss.

After the training for every sub-model is done, we discard their linear classifier, apply a projection layer to features from time-series encoder and text encoder to make the size of the outputs the size of features from image encoder). This is done to stop the model from overfitting to one modality. We then concatenate these latent hidden features, giving us a feature vector the size of $512 + 512 + 512 = 1536$. Different ways have been proposed to aggregate modalities. Clinical variables can be fused with other extracted features by simple concatenation (Spasov et al., 2018; Yala et al., 2019), replicated and scaled followed by concatenation (Youngjin Yoo and Tam, 2019) or by getting maximum from every modality output feature Vale-Silva and Rohr (2021). We will use simple concatenation but will experiment with this in future work. We then apply to the concatenated feature either a 1) simple linear layer to predict the outputs or 2) a single LSTM Layer with a hidden dimension size of 512 like Hayat et al. (2023). Using a LSTM for prediction mirrors how doctors make decisions, gradually processing information from each modality one at a time. This allows the LSTM to initially learn from text data and then, guided by that initial understanding, incorporate chest X-ray information to refine its internal state. This sequential learning also enables the model to handle situations where data from one modality is missing. We then apply a linear classifier on the output of the LSTM to predict the lables. After this we train all the model in an end-to-end fashion.

Different studies proposed a multimodal dropout method where at training time, the data from the missing modalities are replaced by zero with a predetermined probability p, but none use the same mechanism as inverse dropout method where the rest of the data is multiplied to accommodate the missing data. We will continue with the established practice with the minor divergence of multiplying by $\frac{1}{1-p}$ with $p = 0.3$. Though are results showed mixed improvements on performance.

We train the models for two tasks:

- **In-hospital mortality prediction:** Mortality prediction is one of the primary outcomes of interest of an hospital admission. We formulate mortality as a binary classification task, where the label indicates the death event for a patient in the first 48 hours spent in the ICU.

- **Phenotype classification/ICD code group prediction:** In this multi-label classification problem, the goal is to predict the presence or absence of 25 diverse medical conditions (chronic, mixed, and acute) for each patient admitted to the ICU. Becuase MIMIC-IV contains both ICD-9 and ICD-10 codes, we map all ICD-10 codes to ICD-9 for uniformity.

We preprocess and split the data like (Hayat et al., 2023) to facilitate comparison. We performed no hyperparamter search and used the same hyperparameters as (Hayat et al., 2023). Both tasks are evaluated using AUROC and AUPRC. For clinical notes we use both discharge notes and radiology notes and for chest X-ray images we only use the front view.

4 Results

To compare benchmarks, we only use the best results from Hayat et al. (2023) and discard results for other configurations. In the data, the text modality and EHR modality are always present for every patient but there are samples where the chest X-ray images are absent. We train data on two datasets, one where all modalities are present in all samples and one where EHR and text are present but image modality is not present (in all samples). The results for the first scenario will be known as multimodal without dropout and the other one will be known as multimodal with dropout. The reason for this naming is that in the second scenario we use the inverted dropout method explained in the previous section to accommodate the missing data. The unimodal models are trained on all the available data for that modality.

4.1 Results for in-hospital mortality prediction

Table 2 shows the comparison between the results published in Hayat et al. (2023) and the results obtained by applying our model for for in-hospital mortality prediction. The results are self explanatory: chest X-ray images are the modality with the most amount of information and after that is the text modality. Although the gap between text and image is small and they have comparable performance. The clinical data have the least information resulting in the lowest benchmarks. The last row which is our mutlimodal approach outperforms both unimodal approaches and also the results from Hayat et al. (2023) though still with very small gaps. Here the model without dropout performs slightly better than the model without dropout.

Table 2: Results for 48h mortality prediction in ICU. MedFuse is the results from Hayat et al. (2023). Best results are shown in bold.

Model	AUROC	AUPRC
Only EHR	0.818	0.460
Only text	0.852	0.501
Only image	0.868	0.558
MedFuse	0.874	0.567
Multimodal with dropout	**0.882**	**0.607**
Multimodal without dropout	**0.883**	**0.609**

4.2 Phenotype classification

Table 3 shows the comparison between the results published in Hayat et al. (2023) and the results obtained by applying our model for for phenotype classification. The pattern for mortality prediction is repeated here. The only difference is the fact that in this task the results are even closer and that the model with dropout performs better. The explaination for results being close is that in multimodal applications, typically one modality is the dominant modality and has the most amount of information while other modalities only have complementary knowledge. This

results in the model overfitting to the dominant modality. It is clear that for this dataset the dominant modality is the chest X-ray images.

Table 3: Results for ICD-9 code group prediction. MedFuse is the results from Hayat et al. (2023). Best results are shown in bold.

Model	AUROC	AUPRC
Only EHR	0.716	0.407
Only text	0.749	0.466
Only image	0.755	0.476
MedFuse	0.770	0.481
Multimodal with dropout	**0.772**	**0.490**
Multimodal without dropout	**0.768**	**0.482**

5 Discussion

5.1 Limitations

This study is not without its limitations. To begin with, we only use one data source which limits generalization. Furthermore, our research could gain significant insights through conducting instance-level analysis. Nonetheless, this necessitates clinical expertise capable of bridging the realms of chest X-ray imaging and clinical time-series analysis, an expertise that is presently lacking within our team.

5.2 Conclusions and Future Work

In this paper, we presented results of several multimodal deep learning models, and ICU scoring systems on two clinical prediction tasks using the publicly available MIMIC-IV datasets. We demonstrated that this framework can outperform similar approaches using less modalities across two healthcare demonstrations, including ICD-9 code group prediction, and 48h mortality prediction. Due to incorporating mechanisms to handle missing data, our models are robust to missing data in test set and the performance does not drop off.

In future, like Ma et al. (2022) we would like to test how a model trained on a multimodal data where the rate of missing data is varied between train and test set, will perform. There are also other modalities such as echocardiogram and electrocardiogram and other free-text notes that we did not use that can be considered in future work. Pre-training the networks in a self supervised manner using all the data which also helps us with missing data problem is also a promising future direction. Applying the methodologies of works like Akbari et al. (2021); Zhang et al. (2023) where they used one model for all modalities and share the model can also be explored.

Declarations

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Data availability

The dataset used during the current study is publicly available from PhysioBank (2000). To access the MIMIC dataset, one should sign up on https://physionet.org/, agree to the data use agreement, complete CITI data or specimens only research training and then download or use the data over the cloud. All the code for the current study was written in Python programming language and will soon be made available on author's github page.

Appendices

Appendix A How to run the code

The file requirement.txt contains the name and version of all the packages used. Follow README.md for instruction on how to run the code.

Appendix B Timeline

Statement of work (one-year effort)

Research phase	Objectives	Deadline
1. Start	Meet with supervisor for initial discussion Read and analyze relevant literature Find data	Esfand 1401
2. Data collection and preparation	Get data Learn data's structure	Ordibehesht 1402
3. Background research and literature review	Find literature Read literature Analyze literature and find gaps Formulate research question	Shahrivar 1402
4. Data analysis	Clean Data Apply method	Azar 1402
5. Writing	Complete a draft Meet with supervisor	Bahman 1402

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